
Energy & Utility Consumption Standard — Bioethanol Plant (excerpt)

Supplied grounding document for capstone groups (e.g. an energy-optimization agent) that don't have their own. Covers the D-201 distillation train and its main utility consumers.

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1. Purpose & scope

Defines target and alarm-level specific energy consumption (SEC) for the bioethanol distillation train, so that deviations can be caught early and investigated before they become a sustained energy loss.

2. Key energy consumers

Consumer	Typical duty	Notes
Reboiler E-201 (LP steam)	~2.4 t/h at normal reflux (L/D = 3.0)	Dominant energy user on the train — see SOP-D201-014
Feed preheat train	Heats beer feed to 85 °C before D-201	Recovers heat from hot bottoms where possible
Feed pump P-450	~12 m ³ /h duty point	Electrical, minor relative to steam
Cooling water pumps (to C-201)	Sized for full condenser duty	See Water Treatment & Cooling SOP

3. Specific energy consumption (SEC) targets

Metric	Target	Alarm (investigate)	Notes
Steam use per litre distillate	2.6 kg steam / L EtOH	> 3.1 kg/L sustained over a shift	Rises sharply if reflux is run high (see §4)
Reflux ratio (L/D)	3.0 (SOP-D201-014 normal)	Sustained > 3.5 without a purity reason	Excess reflux burns steam for no quality benefit
Electrical energy per litre	0.05 kWh/L	> 0.07 kWh/L sustained	Mainly pumps and instrumentation

4. Common energy-loss patterns

1. **Reflux creep** — operators nudging reflux up “to be safe” on purity without confirming AT-207 needs it. Costs steam for no quality benefit. Trim reflux back toward 2.5–3.0 while holding purity ≥ 94% v/v (see SOP-D201-014 §8).
2. **Steam leaks / trap failures** — a step-change rise in reboiler steam flow at constant reflux and feed rate, with bottoms temperature above setpoint, indicates a fouled reboiler or failed steam trap, not a genuine process need.
3. **Cold feed** — feed preheat train underperforming pushes reboiler duty up to compensate; check feed temperature against the 80–92 °C safe range (SOP-D201-014 §3) before adjusting reflux or steam.
4. **Over-cooling** — condenser cooling water flow well above design duty wastes pumping energy without a process benefit; check against cooling water SOP targets.

5. Escalation & human sign-off

- Recommending a reflux or steam-flow change to recover energy is **advisory only**. Any actual setpoint change on D-201 (reflux ratio, reboiler steam) requires shift-engineer authorisation per SOP-D201-014 §3 and §9.
- Do not recommend operating outside the safe ranges in SOP-D201-014 or the Column Setpoints Datasheet (DS-D201-02) to save energy — quality and safety limits take priority over energy savings.
- Sustained SEC alarms (§3) should be logged and escalated to the shift engineer, not resolved automatically.